

Emerging paradigms and new models in urban air mobility, last-mile delivery, and fleet sizing

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EURO WISDOM Forum YoungWomen4OR Talks,
“OR for Routing and Transportation” – October 02, 2025

Outline

Urban Air Mobility



Exact methods

Last-mile Delivery



Mixed Integer Linear
Programming

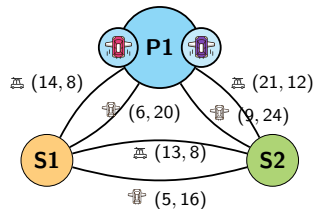
Fleet Sizing



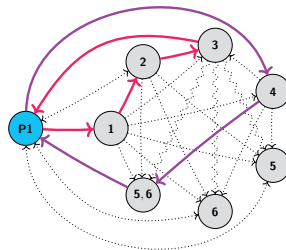
Approximate Dynamic
Programming &
Robust Optimization

The Multimodal Urban Air Mobility Problem with Synchronization Constraints – Problem Definition

Joint work with Feilong Wang (Beihang University) and Roberto Roberti (University of Padova)



- Requests, eVTOLs, vertiplaces
- Payload-dependent energy consumption and battery swap
- Temporal constraints: ready and due times, separation time, safety time, **synchronization**
- **Goal:** maximize the number of passengers served



Two Planes Collide on LaGuardia Airport Taxiway

A plane that was preparing to take off struck a plane that had landed, the authorities said. One passenger was injured.



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Delta Airlines planes at La Guardia Airport in New York in 2023. Ed Jones/Agence France-Presse — Getty Images



By [Francesca Regalado](#) and [Mark Walker](#)

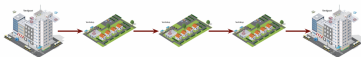
Oct. 2, 2025 Updated 3:35 a.m. ET

The Multimodal Urban Air Mobility Problem with Synchronization Constraints – Methodology and Results

1 (Not-so) Compact model using **arcs**

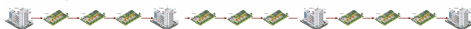


2 Integer relaxation using **trips**

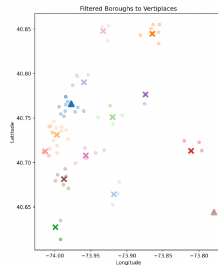


+ Synchronization with **Flexible Job Shop Scheduling** → **Branch-and-Cut**

3 Integer relaxation using **journeys**



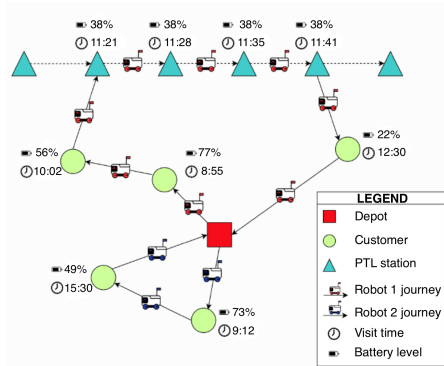
+ Synchronization with **Flexible Job Shop Scheduling** → **Branch-and-Price-and-Cut**



- Realistic instances from NYC taxi data
- BC solves instances up to 30 requests (1h), 10 (1.5h), and 10 (2h)
- BPC closes all 40 (1h), 40 (1.5h), and 30 (2h); largest: 60 requests

Last-mile Delivery with Autonomous Robots and Public Transport Lines – Problem Definition

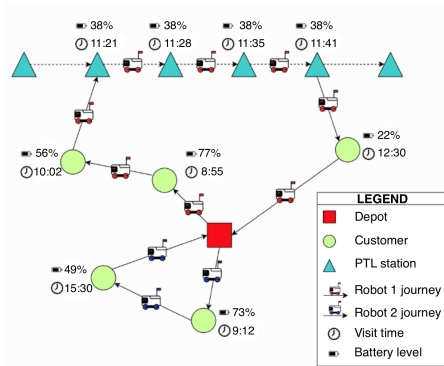
Joint work with Antonino A. Carè (Politecnico di Milano), Roberto Roberti (University of Padova), Sara Stoia (University of Calabria), and Yanlu Zhao (Durham University)



- Single depot, homogeneous robots, customers, and **metro stations**
- Payload-dependent energy consumption and battery swap
- **Goal:** minimize total cost to serve customers

Last-mile Delivery with Autonomous Robots and Public Transport Lines – Problem Definition

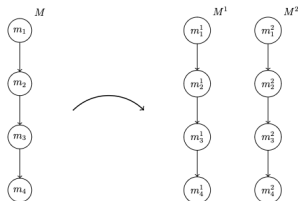
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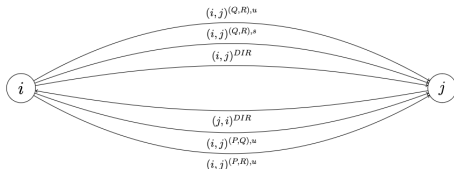
Last-mile Delivery with Autonomous Robots and Public Transport Lines – Methodology and Results

1 Shift network:

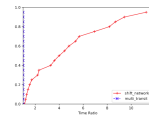
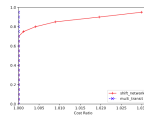
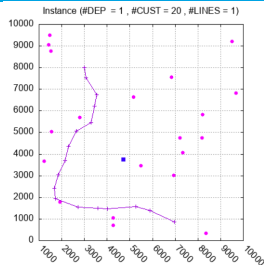


→ Traditional x_{ij}

2 Multi-transit network:



→ x_{ij}^{DIR} and $x_{ij}^{PR,s}$ + dominance rule



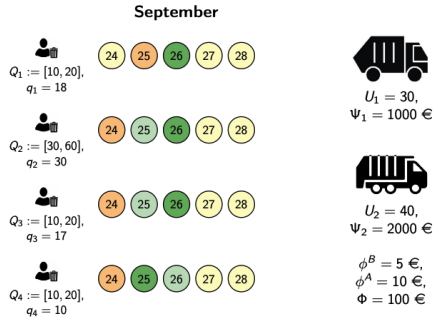
- Realistic instances with 20, 40, or 60 customers and 2, 4, or 8 robots
- Sensitivity analyses

Robust Policies for a Multi-Period Fleet Sizing Problem under Demand Uncertainty – Problem Definition

Joint work with Demetrio Laganà (University of Calabria) and Roberto Roberti (University of Padova)



- Stochastic-dynamic setting over multiple days
- Customers with **uncertain demand in known intervals**, day windows, and penalties
- Vehicles with $\neq$ capacities and fixed costs
- **Goal:** minimize total costs due to fleet sizing, staffing, and customer inconvenience



Optimal solution (value = 2205 €) –
Employ vehicle 2 to serve:

- Customers 1 and 3 on 09/26
- Customers 2 and 4 on 09/25

Robust Policies for a Multi-Period Fleet Sizing Problem under Demand Uncertainty – Methodology

Lower bounds:

- 1 EP – Deterministic ex-post:
 - Variables for service, usage, and number of vehicles of each type each day
 - Constraints on service, capacity, vehicle availability
- 2 QMIN – Pending orders in the objective function and uncertain orders with minimum demand in constraints

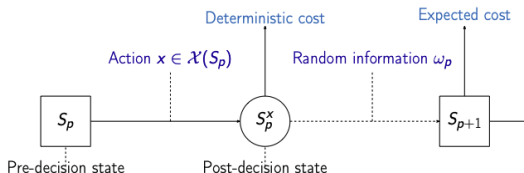
Upper bounds:

- 1 WC – Worst case (maximum demand)
- 2 RO – Robust counterpart: **budget of uncertainty** and cardinality-constrained uncertainty set

$$\sum_{i \in N_0} q'_i a_{ikp} + \sum_{i \in \tilde{N}} \underline{q}_i a_{ikp} + \max_{\substack{I \subseteq \tilde{N} \\ |I| \leq \Gamma}} \sum_{i \in S} (\bar{q}_i - \underline{q}_i) a_{ikp} \leq R_k u_{kp} \quad k \in K, p \in P \setminus \{0\}$$

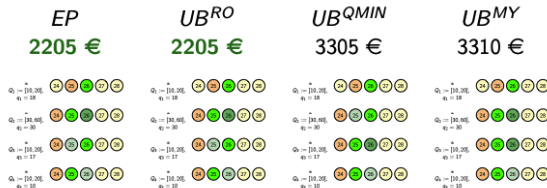
Linearization through dualization or recursive equations

Robust Policies for a Multi-Period Fleet Sizing Problem under Demand Uncertainty – Methodology



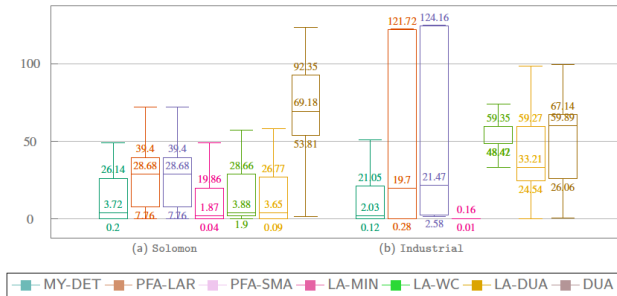
Definition of **Approximate Dynamic Programming** policies applied in a folding-horizon fashion to provide **upper bounds**

$$QMIN \leq EP \leq RO, \text{UB} \leq WC$$



Robust Policies for a Multi-Period Fleet Sizing Problem under Demand Uncertainty – Results

- Tests on Solomon instances (25 or 50 customers, homogeneous vehicles) and others derived from an industrial case study (≈ 200 customers, small and large vehicles)
- Policy performance and price of robustness



- Sensitivity analyses increasing the degree of uncertainty, on extreme realizations, and varying maximum demand

References – Urban Air Mobility



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References – Fleet Sizing



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Thank you for your attention!
Any question or feedback is welcome.

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